



Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology

Course Code & Title : IT3501 Full Stack Web Development

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 5 / Basic React Application
- **Course Outcome** : CO5
- **Topic Learning Outcome:** TLO21
- **Activity Chosen** : Learning by Teaching
- **Justification:**
 - Learning by Teaching activity is chosen to make the students to learn about how to develop a basic React application to display student details. By this activity, Students will learn the assigned topic individually and teach the topic to their peers, which improve their communication, presentation and self-learning skills.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 - Learning by teaching is a mode of learning where students are expected to comprehend material and then prepare presentations to teach their peers.
 - The topic Basic React Application was given to the student Ms.S.Sobana one week before the presentation.
 - The student S.Sobana explained the steps involved in developing a basic react application to display student details on board presentation to their class mates.
 - The topics were delivered around 30 minutes. Finally doubts in the corresponding topic was clarified and summarized.
 - The implementation of Learning by teaching activity as shown in Figure 1 and Figure 2.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO5	3	2	3	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	The Students will be able to understand the steps involved in developing a simple react applications	The Students will be able to develop simple React application using create-react-app tool	The Students will be able to design and develop interactive web application using React	As the student is taking the class, his/her individuality can be improved and able to develop React applications individually	The Students presentation skill will be improved as they are facing the audience and teaching the concepts	The Students will be to apply the React concepts in developing interactive front-end web applications

- **Images / Screenshot of the practice:**



Figure 1: Screenshot of S.Sobana Learning by teaching activity presentation

- **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

- All the Students were eagerly listened the class and they understand the concept of developing a basic react application to display student details
- Ms.S.Sobana expressed that, the activity helps in improving self-learning and presentation skills.
- They felt comfortable to ask doubts while teaching the topic.

❖ ***Benefit of the practice:***

- Learning by teaching activity improves the technical knowledge, self-learning and communication among students.
- This peer-peer learning activity is very interactive and made the slow learners to clarify the doubts with their peers.
- The Students can share their gained knowledge.

❖ ***Challenges faced in implementation:***

- As the activity involves only one student, unable to get all students in self-learning.
- Need to demonstrate the React concept again to all students.
- The activity needs more time to share their views

Also, I had demonstrated the steps involved in developing basic react application with an example.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_VA_CS8603_Learning_teaching.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/6_GS_CS8592_learning_by_teaching.pdf

Signature of Faculty Member

HOD